Summary of Proposed Feature Set

This table details the acoustic, spectral, phonetic, and vocal technique descriptors used in the proposed framework, along with their significance in the field of singing analysis.

Category / Descriptors	Feature Name	Description and Relevance
PROSODIC DESCRIPTORS		
Fundamental Frequency Statistics	`F0semitoneFrom27.5Hz_sma3nz_{amean, stddevNorm, percentile20.0, percentile50.0, percentile80.0, pctlrange0-2, meanRisingSlope, stddevRisingSlope, meanFallingSlope, stddevFallingSlope}, pitch_variability, pitch_bias`	Tracks pitch range, average register, and the speed and variability of pitch transitions.
Intonation	`dett_{mean, std, cv}`	Quantifies micro-deviation from the nearest semitone, serving as a proxy for tuning accuracy and pitch stability
Delta Pitch & Melodic Deviation	`delta_pitch_{mean, std}, melodic_deviation`	Captures short-term pitch dynamics (pitch drift) and the overall melodic variability of the performance.
Jitter & Shimmer	`jitterLocal_sma3nz_{amean, stddevNorm}, shimmerLocaldB_sma3nz_{amean, stddevNorm}`	Measures micro-variations in pitch and volume to detect vocal instability or irregular phonation.
Loudness	`loudness_sma3_{amean, stddevNorm, percentile20.0, percentile50.0, percentile80.0, pctlrange0-2, meanRisingSlope, stddevRisingSlope, meanFallingSlope, stddevFallingSlope}, equivalentSoundLevel_dBp, delta_energy_{mean, std}, {st, lt}_loudness_variability`	Tracks perceived volume, overall dynamic range, short-term and long-term loudness variability and the sharpness of vocal attacks and decays.
Loudness Peak Rate	`loudnessPeaksPerSec`	Measures the rate of loudness fluctuations
Voiced/Unvoiced Distribution	`VoicedSegmentsPerSec, {Mean, Stddev}VoicedSegmentLengthSec, {Mean, Stddev}UnvoicedSegmentLength`	Quantifies the proportion of voiced versus unvoiced frames and the presence of silent or non-voiced segments.
SPECTRAL DESCRIPTORS		
Harmonics-to-Noise Ratio (HNR)	`HNRdBACF_sma3nz_{amean, stddevNorm}`	Measures the ratio of harmonic to noise components of the signal, reflecting vocal clarity and distinguishing clean phonation from breathiness/raspiness.
Formants	`F1{frequency, bandwidth, amplitudeLogRelF0}_sma3nz_{amean, stddevNorm}, F2{frequency, bandwidth, amplitudeLogRelF0}_sma3nz_{amean, stddevNorm}, F3{frequency, bandwidth, amplitudeLogRelF0}_sma3nz_{amean, stddevNorm}`	Captures vocal tract resonances associated with vowel articulation and overall tone quality.
Alpha Ratio & Hammarberg Index	`alphaRatioV_sma3nz_{amean, stddevNorm}, alphaRatioUV_sma3nz_amean,`	Measures the spectral balance between low and high frequency energy, providing an indicator of

Category / Descriptors	Feature Name	Description and Relevance
	hammarbergIndexV_sma3nz_{amean, stddevNorm}, hammarbergIndexUV_sma3nz_amean	vocal brightness, tension, and breathiness.
Harmonic Difference	logRelF0-H1-H2_sma3nz_{amean, stddevNorm}, logRelF0-H1-A3_sma3nz_{amean, stddevNorm}	Quantifies relationships between harmonic components of the signal, providing information related to voice quality.
Spectral Slope	slopeV0-500_sma3nz_{amean, stddevNorm}, slopeV500-1500_sma3nz_{amean, stddevNorm}, slopeUV0-500_sma3nz_amean, slopeUV500-1500_sma3nz_amean	Quantifies the rate at which spectral energy decreases as frequency increases, reflecting vocal brightness and spectral tilt.
Spectral Flux	spectralFlux_sma3_{amean, stddevNorm}, spectralFluxV_sma3nz_{amean, stddevNorm}, spectralFluxUV_sma3nz_amean	Measures frame-to-frame changes in the signal's spectrum, capturing temporal variation of perceived timbre.
Spectral Centroid & Flatness	spectral_centroid_{mean, std}, spectral_flatness_{mean, std}	Represents the center of mass and uniformity of the spectrum, correlating with perceived brightness and noise distribution.
Zero Crossing Rate	zcr_{mean, std}	Helps differentiate noisy, unvoiced, or breathy sounds from clear, periodic singing.
VOCAL TECHNIQUE DESCRIPTORS		
Vocal Technique	Tech_{mean, std, max, peak_factor}	Tracks the presence of specific singing techniques (Falsetto, Vibrato, Mix, Breathy, Glissando, Pharyngeal).
Modulation Frequency Domain Ratios	MFDR_{slow, vibrato, high, slow2vibrato, vibrato2high}	Analyzes spectral modulation of the pitch across specific frequency bands: slow (0.5–3 Hz, portamento/scoops), vibrato (4–8 Hz, sustained vibrato), and high (8–20 Hz, involuntary vocal tremor)
Technique Entropy	Tech_entropy	Shannon entropy over technique probabilities. Evaluates whether a singer uses stable vocal mode or rapidly shifts through a complex vocal palette.
Dominant Technique Changes	Num_tech_changes	Serves as an indicator of vocal agility by measuring the count of switches among dominant predicted vocal modes
PHONETIC DESCRIPTORS		
Articulation Rate	Articulation_rate	Measures the rate of phoneme production, excluding pause intervals.
Vowel & Pause Ratios	Vowel_ratio, pause_ratio	Measures how much of the performance is continuous singing (vowels) and how much is gaps in the vocal delivery (pauses).
Pause Intervals & Distribution	{mean, std}_pause_duration, avg_distance_pauses	Analyzes breath control and phrasing.
Phoneme Duration	{mean, std}_phoneme_duration, phoneme_duration_cv	Evaluates delivery by analyzing the stretch and variability of vowels and consonants.